# Fundamentals of Statistical Learning

Definitions and results cheatsheet

## Chapter 1. Defining probability

**Definition 1.1: Outcomes and events**

A random experiment is a process whose result is not known in advance. Each possible result is an *outcome* $\omega$. The set of all possible outcomes is the *sample space* $\Omega$. An *event* is a subset $A \subseteq \Omega$; it occurs when the observed outcome belongs to $A$.

**Proposition 1.1: De Morgan's laws**

For any events $A$ and $B$,

$$(A \cup B)^c = A^c \cap B^c, \qquad (A \cap B)^c = A^c \cup B^c.$$

More generally, for events $A_1, \dots, A_n$,

$$\left(\bigcup_{i=1}^{n} A_i\right)^c = \bigcap_{i=1}^{n} A_i^c, \qquad \left(\bigcap_{i=1}^{n} A_i\right)^c = \bigcup_{i=1}^{n} A_i^c.$$

**Definition 1.2: Probability on a finite sample space**

A probability measure $P$ assigns a number $P(A)$ to every event $A \subseteq \Omega$ such that:

1. $0 \le P(A) \le 1$ for every event $A$;
2. $P(\Omega) = 1$;
3. if $A \cap B = \varnothing$, then $P(A \cup B) = P(A) + P(B)$.

The last property is called *finite additivity*.

**Proposition 1.2: Inclusion–exclusion for two events**

For any events $A$ and $B$,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

**Proposition 1.3: Union bound (Boole's inequality)**

For events $A_1, \dots, A_n$,

$$P\left(\bigcup_{i=1}^{n} A_i\right) \le \sum_{i=1}^{n} P(A_i).$$

**Proposition 1.4: Bonferroni's inequality**

For events $A_1, \ldots, A_n$,

$$P\left(\bigcap_{i=1}^{n} A_i\right) \geq \sum_{i=1}^{n} P(A_i) - (n-1).$$

# Chapter 2. Conditional probability

**Definition 2.1: Conditional probability**

For events $A, B$ with $P(B) > 0$, the probability of $A$ given $B$ is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

**Definition 2.2: Independent events**

Two events $A$ and $B$ are independent if

$$P(A \cap B) = P(A)P(B).$$

**Theorem 2.1: Bayes' theorem**

For a group $B_j$ of the partition[†] and an event $A$ with $P(A) > 0$,

$$P(B_j \mid A) = \frac{P(A \mid B_j)P(B_j)}{P(A)} = \frac{P(A \mid B_j)P(B_j)}{\sum_{i=1}^{m} P(A \mid B_i)P(B_i)}.$$

# Chapter 3. Random variables

**Definition 3.1: Random object**

A *random object* is a measurable[*] map from outcomes to possible observations:

$$Z : \Omega \longrightarrow \mathcal{T}, \qquad \omega \longmapsto Z(\omega).$$

The set $\mathcal{T}$ says what kind of thing we record. It could contain numbers, vectors, images, or text messages. Once $\omega$ is known, $Z(\omega)$ is the observation we get.

**Definition 3.2: Random variable**

A *random variable* is a measurable map

$$X : \Omega \longrightarrow \mathbb{R}.$$

It assigns one real number $X(\omega)$ to each outcome $\omega$.

[†]Here $B_1, \ldots, B_m$ form a partition of the sample space, with $P(B_i) > 0$.
[*]A rigorous definition of measurability is beyond the scope of this course.

**Definition 3.3: Random vector**

A *p-dimensional random vector* is a measurable map

$$\underline{X} : \Omega \longrightarrow \mathbb{R}^p, \qquad \underline{X}(\omega) := \big(X_1(\omega), \ldots, X_p(\omega)\big).$$

Each coordinate $X_j$ is a real-valued random variable. Thus one outcome gives us $p$ numbers collected into a vector. These measurements can be related; they do not have to be independent.

**Definition 3.4: Preimage**

For a random variable $X : \Omega \to \mathbb{R}$ and a set $E \subseteq \mathbb{R}$, the *preimage* (or *inverse image*) of $E$ is

$$X^{-1}(E) := \{\omega \in \Omega : X(\omega) \in E\} \subseteq \Omega.$$

To find this event, we collect the outcomes whose outputs fall in $E$. Whenever $X^{-1}(E)$ is an event, we define

$$\begin{aligned} P(X \in E) &:= P\big(X^{-1}(E)\big) \\ &= P\big(\{\omega \in \Omega : X(\omega) \in E\}\big). \end{aligned}$$

In particular, for a single value $x$,

$$P(X = x) := P\big(X^{-1}(\{x\})\big).$$

Here $X$ is the random variable, while $x$ is a particular real number.

**Definition 3.5: Distribution of a random variable**

The *distribution* of $X$ is the rule that tells us with what probability $X$ takes values in a given set $E$.

$$P_X(E) = P(X \in E).$$

**Definition 3.6: Discrete random variable and PMF**

A random variable $X$ is *discrete* if there is a finite or countably infinite set[‡] $\mathcal{X} = \{x_1, x_2, \ldots\}$ such that $P(X \in \mathcal{X}) = 1$.

Its *probability mass function* (PMF) is

$$p_X(x) := P(X = x), \qquad x \in \mathbb{R}.$$

1. $p_X(x) \geq 0$ for every $x \in \mathbb{R}$.
2. The *support*, in the discrete sense used here, is $\operatorname{supp}(X) := \{x \in \mathbb{R} : p_X(x) > 0\}$. Thus the support lists the values that have positive probability; values outside it have probability zero.
3. $\displaystyle\sum_{x \in \mathcal{X}} p_X(x) = 1.$

[‡]A finite set has finitely many elements. A set $A$ is countably infinite if there exists a bijection $f : \mathbb{N} \to A$.

**Definition 3.7: Continuous random variable and PDF**

A random variable $X$ is *continuous* (in the sense used here) if there exists a function $f_X$, called its *probability density function* (PDF), such that, for every $a \le b$,

$$P(a \le X \le b) = \int_a^b f_X(x)\,dx.$$

1. $f_X(x) \ge 0$ for every $x \in \mathbb{R}$.
2. Following the convention in the slides, the *support* is $\mathrm{supp}(X) := \{x \in \mathbb{R} : f_X(x) > 0\}$.
3. $\int_{\mathbb{R}} f_X(x)\,dx = 1$.

**Definition 3.8: Cumulative distribution function**

Given a random variable $X$, its *cumulative distribution function* (CDF) is the function $F_X : \mathbb{R} \to [0,1]$ defined by

$$F_X(x) := P(X \le x) = P\big(X \in (-\infty, x]\big), \qquad x \in \mathbb{R}.$$

It gives the probability accumulated up to and including the threshold $x$. It is defined for both discrete and continuous random variables.

**Proposition 3.1: Properties of the CDF (I)**

A function $F(\cdot)$ mapping the real line to $[0,1]$ is a CDF for some probability measure $P(\cdot)$ if and only if:

1. $F(\cdot)$ is *non-decreasing*: $x_1 < x_2$ implies that $F(x_1) \le F(x_2)$.
2. $F(\cdot)$ is *normalized*:
$$\lim_{x \to -\infty} F(x) = 0, \qquad \lim_{x \to +\infty} F(x) = 1.$$
3. $F(\cdot)$ is *right-continuous*: for all $x$ we have
$$F(x) = F(x^+), \qquad F(x^+) := \lim_{t \downarrow x} F(t).$$

**Proposition 3.2: Properties of the CDF (II/a)**

Let $F_X(\cdot)$ be the CDF of a random variable $X$. Then:

1. For $x_1 < x_2$,
$$P(x_1 < X \leq x_2) = P\big(X \in (x_1, x_2]\big) = F_X(x_2) - F_X(x_1).$$
In the discrete case, be careful with the strict and non-strict inequalities.
2. $$P(X > x) = P\big(X \in (x, +\infty)\big) = 1 - F_X(x).$$
3. If $X$ is continuous with PDF $f_X(\cdot)$, then
$$F_X(x) = \int_{-\infty}^{x} f_X(t)\, dt.$$
A PDF can be recovered from the CDF as
$$f_X(x) = F_X'(x)$$
at points where $F_X$ is differentiable.

**Proposition 3.3: Properties of the CDF (II/b)**

Let $F_X(\cdot)$ be the CDF of a random variable $X$. Then:

1. $$P(x_1 < X \leq x_2) = P\big(X \in (x_1, x_2]\big) = F_X(x_2) - F_X(x_1), \qquad x_1 < x_2.$$
2. $$P(X > x) = P\big(X \in (x, +\infty)\big) = 1 - F_X(x).$$
3. If $X$ is discrete with PMF $p_X(\cdot)$ and support $\mathcal{X} = \{x_1, x_2, \ldots\}$, then
$$F_X(x) = \sum_{x_i \leq x} p_X(x_i) = P(X < x) + p_X(x).$$
Hence
$$p_X(x) = F_X(x) - F_X(x^-), \qquad F_X(x^-) := \lim_{t \uparrow x} F_X(t).$$

**Definition 3.9: Quantile function: continuous case**

Let $X$ be a continuous random variable with CDF $F_X$. For a probability level $0 < p < 1$, its *quantile function* is
$$Q(p) := \inf\{x \in \mathbb{R} : F_X(x) = p\}.$$
It gives the smallest value $x$ such that $F_X(x) = P(X \leq x) = p$. We also write $q_p$ instead of $Q(p)$.

**Definition 3.10: Quantile function: general case**

Let $X$ be any random variable, discrete or continuous, with CDF $F_X$. For a probability level $0 < p < 1$, its *quantile function* is

$$Q(p) := \inf\{x \in \mathbb{R} : F_X(x) \geq p\}.$$

It gives the smallest threshold $x$ such that $P(X \leq x)$ is at least $p$. We also write $q_p$ instead of $Q(p)$.

**Definition 3.11: Uniform distribution**

A continuous random variable $X$ has a *uniform distribution* with parameters $a < b$, where $a, b \in \mathbb{R}$, if its PDF and CDF are

$$f_X(x) = \frac{1}{b-a}\mathbf{1}_{[a,b]}(x),$$

$$F_X(x) = \begin{cases} 0, & x < a, \\ \dfrac{x-a}{b-a}, & a \leq x \leq b, \\ 1, & x > b. \end{cases}$$

We denote this distribution by $\mathrm{Unif}(a, b)$ and write $X \sim \mathrm{Unif}(a, b)$. Here Unif abbreviates *uniform*, and $\sim$ means *has distribution*.

**Definition 3.12: Exponential distribution**

We write $T \sim \mathrm{Exp}(\lambda)$, with rate $\lambda > 0$, when its PDF and CDF are

$$f_T(t) = \lambda e^{-\lambda t}\mathbf{1}_{[0,+\infty)}(t), \qquad F_T(t) = \begin{cases} 0, & t < 0, \\ 1 - e^{-\lambda t}, & t \geq 0. \end{cases}$$

**Definition 3.13: Bernoulli distribution**

For $0 \leq p \leq 1$, we write $X \sim \mathrm{Ber}(p)$ when

$$p_X(x) = \begin{cases} 1-p, & x = 0, \\ p, & x = 1, \\ 0, & \text{otherwise.} \end{cases}$$

The parameter $p$ is the success probability. The two masses add to 1.

**Definition 3.14: Binomial distribution**

For an integer $n \geq 1$ and $0 < p < 1$, $X \sim \mathrm{Bin}(n, p)$ has PMF

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, \ldots, n,$$

and zero elsewhere. For $p = 0$, $X = 0$ with probability 1; for $p = 1$, $X = n$ with probability 1.

**Definition 3.15: Beta distribution**

For $\alpha, \beta > 0$, the density of $P \sim \text{Beta}(\alpha, \beta)$ is

$$\pi(p) = \frac{p^{\alpha-1}(1-p)^{\beta-1}}{B(\alpha,\beta)}, \qquad 0 < p < 1,$$

and it is zero elsewhere. The denominator is the normalizing constant

$$B(\alpha,\beta) = \int_0^1 t^{\alpha-1}(1-t)^{\beta-1}\,dt.$$

**Definition 3.16: Gamma distribution**

Using the Gamma function[§] defined above, we write $T \sim \text{Gamma}(a, \lambda)$ when its PDF is

$$f_T(t) = \frac{\lambda^a}{\Gamma(a)} t^{a-1} e^{-\lambda t} \mathbf{1}_{(0,+\infty)}(t).$$

# Chapter 4. Random vectors

**Definition 4.1: Random vector**

On a common probability space $(\Omega, \mathcal{F}, P)$, a **random vector** is a vector of random variables

$$\underline{X} : \Omega \longrightarrow \mathbb{R}^p, \qquad \underline{X}(\omega) = \begin{pmatrix} X_1(\omega) \\ \vdots \\ X_p(\omega) \end{pmatrix}.$$

The integer $p$ is the dimension of the vector. Each component $X_j : \Omega \to \mathbb{R}$ is a random variable.

**Definition 4.2: Independence of random variables**

Two random variables $X$ and $Y$ are independent when

$$P(X \in I,\ Y \in J) = P(X \in I)P(Y \in J) \quad \text{for every pair of intervals } I, J.$$

[§]In the notes, $\Gamma(a) = \int_0^{+\infty} t^{a-1}e^{-t}\,dt$ for $a > 0$.

**Definition 4.3: Joint probability mass function**

The **joint PMF** assigns a probability to each pair of values[¶]:

$$p_{X,Y}(x,y) = P(X = x, Y = y) = P\big(\{X = x\} \cap \{Y = y\}\big).$$

It satisfies

$$p_{X,Y}(x,y) \geq 0, \qquad \sum_{x \in S_X} \sum_{y \in S_Y} p_{X,Y}(x,y) = 1.$$

Some pairs in $S_X \times S_Y$ may have probability zero. The joint PMF completely determines the distribution of $(X, Y)$.

**Proposition 4.1: Marginal PMFs**

The marginal distributions are obtained by summing out the other variable:

$$p_X(x) = \sum_{y \in S_Y} p_{X,Y}(x,y), \qquad p_Y(y) = \sum_{x \in S_X} p_{X,Y}(x,y).$$

**Definition 4.4: Conditional PMFs**

For a value $y$ with $p_Y(y) > 0$,

$$p_{X|Y}(x \mid y) = P(X = x \mid Y = y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}.$$

Similarly, for $x$ with $p_X(x) > 0$,

$$p_{Y|X}(y \mid x) = P(Y = y \mid X = x) = \frac{p_{X,Y}(x,y)}{p_X(x)}.$$

For each fixed conditioning value, the conditional probabilities are nonnegative and sum to 1 over the remaining variable.

**Definition 4.5: Independence of discrete random variables**

Two discrete random variables $X$ and $Y$ are independent if and only if

$$p_{X,Y}(x,y) = p_X(x)p_Y(y) \qquad \text{for every } (x,y) \in S_X \times S_Y.$$

Equivalently,

$$p_{X|Y}(x \mid y) = p_X(x) \quad \text{for every } x \text{ and every } y \text{ with } p_Y(y) > 0.$$

[¶]Here $X$ and $Y$ are discrete, with supports $S_X$ and $S_Y$.

**Definition 4.6: Joint cumulative distribution function**

For any pair of random variables, discrete or otherwise,

$$F_{X,Y}(x,y) = P(X \leq x, Y \leq y), \qquad (x,y) \in \mathbb{R}^2.$$

This is the probability of the lower-left region $(-\infty, x] \times (-\infty, y]$ in the plane.

**Proposition 4.2: Marginal CDFs**

The marginal CDFs are obtained as limits of the joint CDF:

$$F_X(x) = \lim_{y \to +\infty} F_{X,Y}(x,y), \qquad F_Y(y) = \lim_{x \to +\infty} F_{X,Y}(x,y).$$

**Proposition 4.3: $X, Y$ independent $\iff$ joint CDF $= F_X F_Y$**

Let $X$ and $Y$ be **arbitrary random variables**, with CDFs $F_X$ and $F_Y$. They are **independent if and only if** their joint CDF satisfies

$$F_{X,Y}(x,y) = F_X(x) F_Y(y) \qquad \text{for all } x, y \in \mathbb{R}.$$

**Definition 4.7: Joint PDF**

A **joint probability density function** $f_{\underline{X}} : \mathbb{R}^p \to [0, +\infty)$ satisfies

$$\int_{\mathbb{R}^p} f_{\underline{X}}(\underline{x})\, d\underline{x} = 1,$$

and gives the probability of a measurable region $A \subseteq \mathbb{R}^p$ through

$$P(\underline{X} \in A) = \int_A f_{\underline{X}}(\underline{x})\, d\underline{x}.$$

**Definition 4.8: Kernel of a PDF**

A **kernel of a PDF** is a nonnegative function $g(\underline{x})$ proportional to that PDF. It specifies the shape of the density, leaving a positive multiplicative constant to be determined.

To obtain the PDF, compute the total integral

$$c = \int_{\mathbb{R}^p} g(\underline{x})\, d\underline{x}.$$

If $0 < c < +\infty$, then

$$f_{\underline{X}}(\underline{x}) = \frac{g(\underline{x})}{c}$$

is the corresponding PDF. The constant $c$ is the **normalizing constant**.

**Proposition 4.4: Marginal PDFs**

The marginal densities are obtained by integrating out the other variable:

$$f_X(x) = \int_{\mathbb{R}} f_{X,Y}(x,y)\,dy, \qquad f_Y(y) = \int_{\mathbb{R}} f_{X,Y}(x,y)\,dx.$$

**Definition 4.9: Conditional PDFs**

When a joint PDF exists, a conditional density of $X$ given $Y = y$ can be defined, for $f_Y(y) > 0$, by

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}.$$

Similarly, for $f_X(x) > 0$,

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x,y)}{f_X(x)}.$$

The conditioning value is held fixed. Integrating the conditional density over the other variable gives 1.

**Proposition 4.5: $X, Y$ independent $\iff$ joint PDF $= f_X f_Y$**

Suppose $X$ and $Y$ **each have a PDF**, denoted by $f_X$ and $f_Y$. They are **independent if and only if** their joint PDF exists and satisfies

$$f_{X,Y}(x,y) = f_X(x) f_Y(y) \qquad \text{almost everywhere.}$$

**Definition 4.10: Bivariate copula**

A **copula** is a joint CDF of two random variables with standard uniform marginals:

$$C(u,v) = P(U \le u, V \le v), \qquad U, V \sim \text{Unif}(0,1).$$

The variables $U$ and $V$ need not be independent. On $[0,1]^2$, the marginal conditions are $C(u,1) = u$ and $C(1,v) = v$.

**Theorem 4.1: Sklar's theorem: bivariate form**

For any joint CDF $F_{X,Y}$ with marginals $F_X$ and $F_Y$, there exists a copula $C$ such that

$$F_{X,Y}(x,y) = C\big(F_X(x), F_Y(y)\big).$$

If both marginals are continuous, the copula is unique. Otherwise, it is uniquely determined only on $\text{Ran}(F_X) \times \text{Ran}(F_Y)$, the product of the sets of values taken by the marginal CDFs.

Conversely, combining any two marginal CDFs with a copula using this formula produces a joint CDF with those marginals.

**Definition 4.11: Independent and identically distributed sample**

A random sample $(X_1, \ldots, X_n)$ is **IID** from $F_X$ if

- the random variables $X_1, \ldots, X_n$ are mutually independent;
- each $X_i$ has the same CDF $F_X$.

Here $n$ is the sample size. Uppercase $X_i$ denotes a random observation before sampling; lowercase $x_i$ denotes its observed value. The realized dataset is $\mathcal{D}_n = (x_1, \ldots, x_n)$.

**Definition 4.12: Likelihood**

For fixed observed data $\mathcal{D}_n = (x_1, \ldots, x_n)$ and a model parameter $\theta \in \Theta$, the **IID likelihood** is

$$L(\theta; x_1, \ldots, x_n) = \begin{cases} \displaystyle\prod_{i=1}^{n} p_\theta(x_i), & \text{discrete model,} \\ \displaystyle\prod_{i=1}^{n} f_\theta(x_i), & \text{model with a PDF.} \end{cases}$$

**Definition 4.13: Maximum likelihood estimation**

A **maximum likelihood estimator (MLE)** chooses

$$\widehat{\theta}_n \in \arg\max_{\theta \in \Theta} L(\theta; x_1, \ldots, x_n),$$

when a maximizer exists. Here $\widehat{\theta}_n$ is the parameter estimate calculated from the $n$ observations, and arg max denotes the set of parameter values attaining the largest likelihood.

**Definition 4.14: Empirical cumulative distribution function**

Given observations $x_1, \ldots, x_n$, the **empirical CDF** is

$$\widehat{F}_n(t) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}_{\{x_i \le t\}}.$$

It is the fraction of observations at or below $t$. The corresponding empirical distribution assigns mass $1/n$ to each observation, counting repetitions.

# Chapter 5. Stochastic simulations

**Definition 5.1: Pseudorandom number generator**

In a simplified description, a *pseudorandom number generator* (PRNG) uses a deterministic function

$$f : (0,1) \longrightarrow (0,1)$$

and a starting value, or *seed*, $u_0 \in (0,1)$ to generate

$$u_0, \quad u_1 = f(u_0), \quad u_2 = f(f(u_0)), \quad \ldots, \quad u_n = f^n(u_0).$$

The function is chosen so that the sequence imitates the statistical behaviour of an IID Uniform$(0,1)$ sample. Here $f^n(u_0)$ means applying $f$ repeatedly $n$ times.

**Definition 5.2: Linear congruential generator**

A *linear congruential generator* (LCG) uses integer parameters $\alpha$, $\beta$, and $M > 1$, and an integer seed $x_0 \in \{0, 1, \ldots, M-1\}$. Its update rule is

$$x_{k+1} = f(x_k) = (\alpha + \beta x_k) \bmod M.$$

Here $\bmod M$ gives the remainder after division by $M$, so every state $x_k$ belongs to $\{0, 1, \ldots, M-1\}$.

**Proposition 5.1: Quantile transform**

Let $F$ be any CDF and let

$$Q(p) = \inf\{x \in \mathbb{R} : F(x) \geq p\}, \qquad 0 < p < 1,$$

be its quantile function. If $U \sim \text{Unif}(0,1)$ and $Y = Q(U)$, then $Y$ has CDF $F$:

$$F_Y(y) = P(Y \leq y) = F(y), \qquad y \in \mathbb{R}.$$

# Chapter 6. Exploratory data analysis

**Definition 6.1: Class of models**

In this density-estimation problem, a **class of models** $\mathcal{F}$ is a set of candidate PDFs. Each $f \in \mathcal{F}$ represents one possible population distribution.

**Definition 6.2: Penalized log-likelihood**

For fixed observed data $\mathcal{D}_n = (x_1, \ldots, x_n)$ from an IID sample, consider a candidate PDF $f$. Let $J(f) \geq 0$ be its penalty, and let $\lambda > 0$ be the weight given to that penalty. The **penalized log-likelihood** is

$$\log L(f; \mathcal{D}_n) - \lambda J(f) = \sum_{i=1}^{n} \log f(x_i) - \lambda J(f).$$

# Relations

### From the PMF to the CDF

**What we are trying to answer**

If we know the probability of each individual value of $X$, how do we calculate the probability that $X$ is at most $x$?

For a discrete random variable, the event $X \leq x$ includes every possible value $k$ that is at most $x$. These events $\{X = k\}$ are pairwise disjoint, so we add their probabilities:

$$\underbrace{F_X(x)}_{\text{CDF: to find}} = P(X \leq x) = \sum_{\substack{k \in \text{supp}(X) \\ k \leq x}} \underbrace{p_X(k)}_{\text{PMF: known}}, \qquad x \in \mathbb{R}.$$

### From the PDF to the CDF

**What we are trying to answer**

We know how probability is spread over the possible values of $X$ (its PDF). How do we calculate the probability of getting a value less than or equal to $x$?

For a continuous random variable with PDF $f_X$, we calculate $P(X \leq x)$ by integrating the PDF over $(-\infty, x]$:

$$\underbrace{F_X(x)}_{\text{CDF: to find}} = P(X \leq x) = \int_{-\infty}^{x} \underbrace{f_X(t)}_{\text{PDF: known}} \, dt, \qquad x \in \mathbb{R}.$$

### From the CDF to the PMF and PDF

**What we are trying to answer**

We know the CDF $F_X$. How do we recover the PMF $p_X$ if $X$ is discrete, or the PDF $f_X$ if $X$ is continuous?

**What we know.** The CDF is defined by

$$F_X(x) := P(X \leq x).$$

For every threshold $x$, it gives the probability of obtaining a value less than or equal to $x$. We want to recover the PMF or the PDF, depending on the type of $X$.

**Discrete case: recover the PMF.** The PMF is defined by

$$p_X(x) := P(X = x).$$

The CDF includes all values up to $x$. To isolate the probability of the single value $x$, we subtract the probability of values strictly less than $x$:

$$P(X = x) = P(X \leq x) - P(X < x).$$

Since the left limit satisfies

$$F_X(x^-) := \lim_{t \uparrow x} F_X(t) = P(X < x),$$

we obtain

$$\boxed{\underbrace{p_X(x)}_{\text{PMF: to find}} = \underbrace{F_X(x) - F_X(x^-)}_{\text{CDF: known}}.}$$

Thus the PMF at $x$ is the size of the jump of the CDF at $x$.

**Continuous case: recover the PDF.** A PDF is a non-negative function $f_X$ such that

$$P(a \leq X \leq b) = \int_a^b f_X(t)\,dt, \qquad \int_{\mathbb{R}} f_X(t)\,dt = 1.$$

In particular,

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(t)\,dt.$$

The CDF is therefore the integral of the PDF. To recover a PDF, we differentiate the CDF:

$$\boxed{\underbrace{f_X(x)}_{\text{PDF: to find}} = \frac{d}{dx} \underbrace{F_X(x)}_{\text{CDF: known}}.}$$

This recovers a PDF at points where the derivative exists, by the fundamental theorem of calculus.